

Marcello Maugeri

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PROFESSIONAL SUMMARY

I build security testing tools that break complex software systems. My PhD research advances Fuzzing for multiprocess Linux programs, stateful protocols, and GraphQL APIs, and I am now exploring AI-driven approaches to Cybersecurity Governance. In my spare time, I step outside my comfort zone by winning hackathons in AI, DeFi, and Blockchain to test myself against tough challenges.

EDUCATION

PhD in Computer Science (Expected Viva: autumn 2025) Oct 2021 – Sep 2025

University of Catania, Italy.

Dissertation Title: *Automation Challenges and Solutions in Coverage-Guided Fuzzing of Multiprocess Software Systems*.

MSc Computer Science (Major: Cybersecurity and Distributed Systems) Oct 2019 – Sep 2021

University of Catania, Italy

Thesis Title: *Shared Memory Fuzzing*. (Grade 110/110 with honours)

BSc Computer Science (Major: Networks and Cybersecurity) Oct 2016 – Nov 2019

University of Catania, Italy

Thesis Title: *3DCP Dynamic Domotic Device Configuration Protocol and Smart Hub*. (Grade: 110/110 with honours)

RESEARCH AND TEACHING EXPERIENCE

Founding Member and Senior Cybersecurity Research Engineer Jun 2025 – Current

Stealth startup, United Kingdom

- Part of the core team establishing a stealth-stage startup focused on AI-driven cybersecurity and governance solutions for the automotive industry.

Teaching Assistant

Mar 2024 – July 2024

University of Catania, Catania, Italy

- Took in-depth lessons of the following modules: 1) Programming 2 (25 hours) - OOP and Data Structures in C++, Complexity and Sorting Algorithms. 2) Fundamentals of Computer Science (25 hours) - Formal Languages, Computational Models, Logic.
- 100% of the attending students passed the exam on the first attempt.

Research Engineer

Jan 2024 – Jun 2024

University of Catania, Catania, Italy

- Developed the team proposal and won the NGI TrustChain Open Call #2 grant for the PECS project (117,000€).
- Conducted a systematic review of current approaches to Federated Learning in the automotive domain.
- Designed a Federated Learning approach using the Flower framework and released the source code.

Visiting PhD Student

Oct 2023 – Mar 2024

King's College London, London, United Kingdom

- Conducted research on GraphQL security testing, investigating Reinforcement Learning and Fuzzing approaches for enhanced vulnerability discovery.
- Co-designed Wendigo, a Black-Box Deep Reinforcement Learning approach for identifying Denial-of-Service queries in GraphQL APIs.
- Published the paper in DLSP2024, an IEEE S&P workshop.

Project Manager

Mar 2023 – Nov 2024

Helpcode, Genoa, Italy (Remote)

- Conceived and launched an educational platform (<https://app.thewatercode.it/>) that empowers 10–14-year-olds to apply computational thinking to real-world environmental challenges.
- Secured a €279,973 grant, partnering with 10 associations, to fund platform development and outreach.
- Led the team composed of a graphic designer, a storyteller and the development team, monitoring the tasks with Jira.
- Presented to about 30 schoolteachers how to effectively use the platform in their lectures.

Teaching Assistant and Program Coordinator

Nov 2021 – Jun 2023

CINI Cybersecurity Lab UNICT Hub, Catania, Italy

- Coordinated the CyberChallenge.IT program at the University of Catania, mentoring students in Cryptography (2022) and Network Security (2023).
- Managed communication with central organisers and coordinated other local teaching assistants.
- Successfully led the program to enhance students' skills in Cybersecurity fields.

Research Engineer (Internship)

May 2021 – Aug 2021

Robert Bosch GmbH, Renningen, Germany

- Devised an approach based on the fuzzer Honggfuzz and the emulator QEMU to perform fuzzing over programs using shared memory IPC methods.
- Evaluated the approach against the Eclipse Iceoryx IPC layer effectively finding undisclosed bugs.
- Reported the results in my MSc dissertation “Shared Memory Fuzzing”.

Teaching Assistant

Apr 2021 – Jun 2021

University of Catania, Catania, Italy

- Mentored and taught students in Internet Security, fostering critical thinking skills and empowering them to understand fundamental security properties, protocols, firewall configuration, and malware taxonomy.
- Built test environments to perform web security attacks (SQL Injection, XSS), Firewall attacks (misconfigurations, CVE-2016-5410).

Mobile App Engineer

May 2020 – Apr 2021

Codedix, Catania, Italy

- Developed and launched 3 successful apps in app stores using Kotlin, Dart (Flutter), Swift, and Objective-C.
- Utilised Firebase for database and storage capabilities to enhance app performance.
- Collaborated with cross-functional teams to ensure seamless app development and delivery.

Research Intern

Oct 2020 – Dec 2020

Robert Bosch GmbH, Renningen, Germany (Remote)

- Conducted a comprehensive systematic review of 42 state-of-the-art embedded systems fuzzing papers and tools.
- Developed a taxonomy classifying approaches as hardware-based, emulation-based, and abstraction-based.
- Published the paper “Embedded Fuzzing: a review of challenges, tools, and solutions” in the Q1 “Cybersecurity” journal.

Teaching Assistant

Sep 2020 – Oct 2020

University of Catania, Catania, Italy

- Took in-depth lessons of the following modules: 1) Computer Science 101 (30 hours) – Programming Fundamentals, Flowcharts, 2) Programming 2 (30 hours) - OOP and Data Structures in C++, Complexity and Sorting Algorithms.
- Designed and supervised programming exercises (both logical and practical) between courses.

PUBLICATIONS

- *Maugeri, M., Angamo, A., & Bella, G. (2025). KrakQL: LLM-Guided Blind Introspection of GraphQL Schemas. Under review.*
- *Angamo, A., & Maugeri, M. (2025). BenGQL: An Extensible Benchmarking Framework for Automated GraphQL Testing. ASE'25 (Tool Demonstration Track).*
- *Maugeri, M., Castiglione, G., Raciti, M., & Bella, G. (2025). AI-related Vulnerabilities within CVEs: Are We Ready Yet? A Study of Vulnerability Disclosure in AI Products. CCS Workshop (AISec).*
- *Maugeri, M., Daniele, C., & Santoro, F. F. (2025). Fuzzing OPC UA with AFLNet, ChatAFL and LibAFLstar: A Research Experience Paper. IEEE PICom Workshop (HIEMI).*
- *Daniele, C., Bethe, T., Maugeri, M., Continella, A., & Poll, E. (2025). LibAFLstar: Fast and State-aware Protocol Fuzzing. ESORICS.*
- *Castiglione, G., Maugeri, M., & Bella, G. (2025). Poster: Machine Learning for Vulnerability Detection as Target Oracle in Automated Fuzz Driver Generation, DIMVA. doi:10.1007/978-3-031-97620-9_8.*
- *Bella, G., Castiglione, G., Esposito, S., Mangano, M. G., Marchetti, M., Maugeri, M., Raciti, M., Riccobene, S., & Santamaria, D. F. (2025). Privacy Enrooted Car Systems (PECS): Preliminary Design. EAI SmartSP. doi:10.1007/978-3-031-93354-7_17.*
- *Maugeri, M., Morana, M. I. P., Esposito, S., & Bella, G. (2024). User-Empowered Federated Learning in the Automotive Domain. IEEE Blockchain Workshop (Trustchain). doi:10.1109/Blockchain62396.2024.00098.*
- *McFadden, S., Maugeri, M., Hicks, C., Mavroudis, V., & Pierazzi, F. (2024). WENDIGO: Deep Reinforcement Learning for Denial-of-Service Query Discovery in GraphQL. IEEE S&P Workshops (DLSP). doi:10.1109/SPW63631.2024.00012.*
- *Maugeri, M. (2024). Fuzzing Matter(s): A White Paper for Fuzzing the Matter Protocol. ICISSP. doi:10.5220/0012469200003648.*
- *Maugeri, M., Daniele, C., & Bella, G. (2024). Forkfuzz: Leveraging the Fork-Awareness in Coverage-Guided Fuzzing. ESORICS 2023 Workshop (SecAssure). doi:10.1007/978-3-031-54129-2_17*
- *Maugeri, M., Daniele, C., Bella, G., & Poll, E. (2023). Evaluating the Fork-Awareness of Coverage-Guided Fuzzers. ICISSP. doi:10.5220/0011648600003405.*
- *Eisele, M., Maugeri, M., Shriwas, R., Huth, C., & Bella, G. (2022). Embedded fuzzing: a review of challenges, tools, and solutions. Springer Cybersecurity. doi:10.1186/s42400-022-00123-y.*

HONOURS AND AWARDS

- 3rd place, EulerSwap Builder Competition (5,000\$) Jul 2025
- Honourable Mention, Vibe Coding Hack by Encode - Polygon Track (200\$) Jun 2025
- 2nd place, Encode AI London 2025 Hackathon – Starkware AI x DeFi Track (300\$) Apr 2025
- 2 awards at ETHOxford Hackathon - Flare's track 1: pool prize - NERDo Awards (2,000\$) Feb 2025
- 1st place, Encode London 2024 Hackathon - RNDM Agent Track. (5,000\$) Oct 2024
- 1st place tied (2nd place by score time) of CyberChallenge.IT final CTF in Catania. Jun 2021
- 3rd place tied (10th place by age) of "ValorizziAmo Eccellenze" 2015 for the best student in Sicily. May 2015

CERTIFICATIONS

- Github Foundations Jun 2025
- ISC2 Certified in Cybersecurity (CC) Oct 2024
- Google Project Management Specialization Dec 2023
- Google Cybersecurity Specialization Dec 2023
- IELTS Academic (overall 7.5) Apr 2023
- Cisco CCNA: IT Essential, Routing & Switching, Routing and Switching Essentials Jan 2016 – Aug 2016

ADDITIONAL RESPONSIBILITIES

- Peer-reviewed scientific papers for (more recent first): IEEE Transactions on Software Engineering, IEEE PICOM 2025, IEEE Transactions on Industrial Informatics, Information Security Journal: A Global Perspective, IEEE Transactions on Services Computing, COSE Journal, ICECER2024, CCS 2024 (Artifact Evaluation), CMC Journal, ITASEC 2024, ESORICS 2024, ICECET 2024, NeurIPS 2023 Workshop - New in ML, ESORICS 2023, ITASEC 2023.
- Session chair at international conferences: IEEE Cybermatics 2024, ICISSP 2023.
- Supervised 12 BSc and MSc students at the University of Catania and one student at the Royal Institute of Technology (KTH) in Stockholm on research projects and dissertations.

ADDITIONAL TRAINING

Large Language Model Agents MOOC Dec 2024

University of California, Berkeley

Topics: reasoning, planning, tool use, RAG, and ethics of LLM agents.

Fuzzing Summer School

Jun 2024

National University of Singapore, Singapore

Topics: grammar-based fuzzing, coverage-guided fuzzing, database fuzzing, mobile fuzzing.

Cyber in Sophia Antipolis Summer School

Jul 2023

EURECOM, Biot, France

Topics: memory forensics, fuzzing, browser fingerprinting, Bluetooth attacks, kernel exploitation.

Cybersecurity of Internet of Things Solutions

Jun 2023 – Jul 2023

Poznań University of Technology, Poznań, Poland

Topics: data security in IoT solutions, data transmission, data processing and analysis using artificial intelligence.

Winter School on Microarchitectural Security

Dec 2022

Institut Polytechnique de Paris, Paris, France

Topics: side-channel attacks and defences, machine learning, software and hardware microarchitectural.

Agile Management for Industry 4.0

Sep 2022

Poznań University of Technology, Poznań, Poland

Topics: agile management, risk management, business modelling and social competencies.

Cybersecurity Modules (PhD Level)

May 2022 – Jul 2022

University of Genoa, Genoa, Italy

Modules: Mobile Security, IoT Security, and Network Monitoring and Inspection.

Erasmus+

Jan 2021 – May 2021

University College Dublin, Ireland

Modules: Secure Software Engineering, Data Science in Python, Deep Learning, Advances in Wireless Networking.